

predict_spatial_resolution_grid_age.R

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predict_spatial_resolution_grid_age
Predict Spatial-Resolution Grid for One Age

Description

Builds abiotic and optional spatial predictors for one age slice and returns long-format taxon occurrence probabilities.

Usage

```
predict_spatial_resolution_grid_age(  
  prediction_inputs,  
  data_grid,  
  data_grid_coords_projected,  
  age,  
  abiotic_variables,  
  x_lim,  
  y_lim,  
  cache_dir,  
  spatial_mode = "spatiotemporal",  
  climate_fn = NULL,  
  spatial_interpolate_fn = NULL,  
  spatial_basis_project_fn = NULL,  
  spatiotemporal_interpolate_fn = NULL,  
  predict_fn = NULL  
)
```

Arguments

prediction_inputs
Named list returned by [load_spatial_resolution_prediction_inputs()].

<code>data_grid</code>	Prediction grid with 'grid_id', 'coord_long', and 'coord_lat'.
<code>data_grid_coords_projected</code>	Projected prediction grid coordinates with row names 'grid_<id>'.
<code>age</code>	Numeric scalar age in years before present.
<code>abiotic_variables</code>	Character vector of CHELSA variables to extract.
<code>x_lim</code>	Numeric longitude range.
<code>y_lim</code>	Numeric latitude range.
<code>cache_dir</code>	Existing CHELSA cache directory.
<code>spatial_mode</code>	Spatial predictor mode: "spatial", "spatiotemporal", or "none".
<code>climate_fn</code>	Climate extraction function. If 'NULL' (default), [extract_prediction_climate()] is used.
<code>spatial_interpolate_fn</code>	Spatial interpolation function. If 'NULL' (default), [interpolate_mev_to_grid()] is used.
<code>spatial_basis_project_fn</code>	Reusable-basis projection function. If 'NULL' (default), [project_spatial_mev_basis()] is used when basis state is available.
<code>spatiotemporal_interpolate_fn</code>	Spatiotemporal interpolation function. If 'NULL' (default), [interpolate_st_mev_to_grid()] is used.
<code>predict_fn</code>	Prediction function. If 'NULL' (default), the function resolves 'sjSDM::predict.sjSDM'.

Value

Long tibble with predicted probabilities by grid cell, age, and taxon.

Examples

```
## Not run:
predict_spatial_resolution_grid_age(
  prediction_inputs = prediction_inputs,
  data_grid = data_grid,
  data_grid_coords_projected = data_grid_coords_projected,
  age = 0,
  abiotic_variables = c("bio1", "bio12"),
  x_lim = c(-10, 40),
  y_lim = c(35, 70),
  cache_dir = "Data/Temp/chelsa/prediction"
)

## End(Not run)
```

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